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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

C PROGRAMMING PROJECT PRESENTATION

Project Title: [MINI LIBRARY CATALOGUE]

Course: Mini Project

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C PROGRAMMING PROJECT PRESENTATION

INTRODUCTION

* The mini library catalogue is a C-based console application designed to manage a small collection of books efficiently

* It supports adding, displaying, searching, updating, and deleting book records.

* Using file handling, books data is stored permanently tp simulate a small-scale digital library system

OBJECTIVE

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*To understand and implement file handling in C.

* To design and develop a library management system using structures.

*To apply modular and structured programming concepts.

* To maintain book records persistently and efficiently.

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PROBLEM STATEMENT

*Managing a library manually becomes difficult as the number of books increases.

*Traditional record-keeping methods may cause data loss or errors.

*This project provides a simple and reliable console-based system to manage book records effectively using file storage.

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SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 2 GB
- Storage: 100 MB free space

Software Requirements:

- OS: Windows / Linux
- Compiler: GCC or Turbo C / Code::Blocks

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SYSTEM DESIGN

The system follows a modular design.

Modules:

- Input Module – Takes user details for books.
- Processing Module – Performs operations (add, search, delete, etc.)
- File Handling Module – Stores and retrieves records from files.
- Output Module – Displays data on the console.

Flowchart:

User Menu Operation File Storage Display Results

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MODULE DESCRIPTION

1. Add Book – Adds a new book record.
2. Display Books – Shows all stored records.
3. Search Book – Finds a book by ID.
4. Delete Book – Removes a book entry.
5. Update Book – Edits book details.
6. Count Books – Displays total number of books.
7. List by Author – Shows books written by a specific author.
8. List by Year – Lists books published after a given year.

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ALGORITHM / FLOWCHART

Algorithm steps:

1. Start
2. Display Menu
3. Take user choice
4. Perform respective operation
5. Repeat until user exits
6. Stop

(You can insert a flowchart image here in your presentation)

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IMPLEMENTATION

Techniques Used:

- Structures for storing book data
- File handling (fopen, fwrite, fread, etc.)
- Modular programming using functions
- Menu-driven user interface

Sample code:

```
void addBook() { ... }
```

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OUTPUT SCREENS

Sample Console Output:

```
========== MINI LIBRARY CATALOGUE ==========
1. Add a New Book
2. Display All Books
3. Search Book by ID
4. Delete Book by ID
5. Update Book Details
6. Count Total Books
7. Exit
=============================================
```

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RESULTS & DISCUSSION

- The program successfully performs all file-based operations.
- Data remains stored permanently even after program termination.
- The system is user-friendly and efficient for managing small libraries.
- Challenges included handling file updates and avoiding data duplication.

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CONCLUSION & FUTURE ENHANC

Conclusion:

- The project achieved its intended objectives.
- It provided a deep understanding of file handling and data management in C.

Future Enhancements:

- Add a login system for librarians.
- Include categories/genres.
- Generate reports in CSV or text format.
- Create GUI or web interface.

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REFERENCES / ACKNOWLEDGEMENT

References:

- Let Us C – Yashavant Kanetkar
- C Programming – E. Balagurusamy
- Online Tutorials and Documentation

Acknowledgement:

I would like to express my heartfelt thanks to my mentor, department, and college for their support and guidance.

THANK YOU

- “Knowledge is the treasure, and learning is the key.”
- Questions & Discussions